

VINEET JETHANI

+91-7600446877 | Surat, Gujarat | heyvineetjethani@gmail.com | [LinkedIn](#)

EDUCATION

Sardar Vallabhbhai National Institute Of Technology

Surat, Gujarat

B.Tech in Electronics and Communication Engineering - 8.10 CGPA

Aug 2023 - Expected July 2027

Senior Secondary (CBSE)

Surat, Gujarat

Score - 93%

2023

Secondary (CBSE)

Surat, Gujarat

Score - 96.33%

2021

ACADEMIC PROJECTS

EEG-Based Seizure Detection using Spiking Neural Networks

- Designed a hybrid CNN-SNN architecture for EEG-based seizure detection with spike-driven gating mechanism.
- Implemented multi-layer 1D CNN blocks for temporal feature extraction from EEG signals.
- Implemented Leaky Integrate-and-Fire (LIF) neuron dynamics to generate binary spike outputs based on membrane potential thresholding.
- Designed a hard/soft gating mechanism where SNN spike output modulates CNN feature flow before MLP classification.

UART Transmitter

[GitHub Repository](#)

- Designed and implemented a UART Transmitter from scratch in Verilog, focusing on bit level timing and protocol correctness rather than using prebuilt IPs.
- Implemented an internal baud rate generator derived from the system clock to ensure accurate and stable bit timing.
- Designed an FSM based control path with 4 stages: (IDLE, START, DATA, STOP) for reliable transmission.
- Implemented 8N1 UART framing with LSB first data serialization using a shift register datapath.

2G GSM Mobile Signal Jammer

- Designed and analyzed a 900 MHz RF interference system using an NE555 timer and RF transistor (BF495).
- Implemented an LC tank oscillator with tunable frequency using air-core inductors and trimmer capacitors.
- Studied the effects of controlled RF interference on GSM communication for educational and analytical purposes.
- Gained practical understanding of RF oscillators, frequency stability, modulation, and electromagnetic interference (EMI).
- **Report:** [Project Report](#)

SKILLS

Area of Interest:

VLSI Design, Digital Systems, Biomedical Signal Processing

Core Subjects:

Digital Electronics, Signals and Systems

Hardware Description Languages:

Verilog, VHDL

Simulation Tools:

NGspice, LTspice, KiCAD, Vivado, Cadence Virtuoso

Programming:

Python, C (basic)

CERTIFICATIONS

Google Developer Group – SVNIT: Cloud Study Jam 2024

Jan 2025

Explored foundational concepts in cloud computing, AI/ML services, and Google Cloud tools through hands-on labs.